Solving Equations

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KEY WORDS equation *(whārite)* · solve *(whakaoti)* · solution *(otinga)* · inverse operation *(mahi kōaro)*

A One-step equations ●○○

Remember: Undo what has been done to x, and do the same to both sides. To undo $+7$, subtract 7. To undo $\times 4$, divide by 4.

1. Solve.

a) $x + 7 = 12 \quad x =$ _____ **b)** $x - 3 = 6 \quad x =$ _____ **c)** $4x = 24 \quad x =$ _____

d) $\frac{x}{3} = 4 \quad x =$ _____ **e)** $x + 9 = 5 \quad x =$ _____ **f)** $x - 2 = -7 \quad x =$ _____

g) $5x = -15 \quad x =$ _____ **h)** $\frac{x}{2} = -4 \quad x =$ _____ **i)** $4x = 10 \quad x =$ _____

j) $x + 1.5 = 4.5 \quad x =$ _____ **k)** $-3x = -12 \quad x =$ _____ **l)** $x - 0.5 = 1.5 \quad x =$ _____

B Two-step equations ●●○

Remember: Undo the adding or subtracting first, then undo the multiplying or dividing.

2. Solve. Show each step.

a) $2x + 3 = 11$ **b)** $3x - 5 = 13$ **c)** $5x + 1 = 36$ **d)** $4x + 7 = -1$

e) $6x - 4 = 26$ **f)** $2x + 9 = 3$ **g)** $3x + 8 = 8$ **h)** $\frac{x}{3} + 2 = 6$

i) $\frac{x}{2} - 1 = 4$ **j)** $15 - 2x = 7$ **k)** $10x - 7 = 13$ **l)** $8x + 5 = -3$

C Fraction and decimal answers ●●●

Watch out: The answer does not have to be a whole number. If $2x = 5$, then $x = \frac{5}{2}$, which is 2.5.

3. Solve. Give each answer as a fraction in its simplest form.

a) $3x + 1 = 5$ **b)** $2x + 5 = 8$ **c)** $5x - 2 = 2$ **d)** $4x + 3 = 8$

4. Solve. Give each answer as a decimal.

a) $2x + 1 = 6$ **b)** $4x + 3 = 9$ **c)** $5x - 2 = 4$ **d)** $10x + 3 = 10$

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D Checking a solution ●●○

> **Remember:** Substitute the value into the equation. If both sides come to the same number, it is a solution.

5. Is the value of x a solution of the equation? Show your working.

a) $x = 3$ and $2x + 5 = 11$

b) $x = 5$ and $3x - 4 = 10$

c) $x = 4$ and $5x + 1 = 21$

d) $x = 10$ and $\frac{x}{2} + 3 = 8$

e) $x = 4$ and $4x - 7 = 2x + 1$

f) $x = -2$ and $10 - 3x = 4$

E Equations from words ●●●

6. Write an equation for each problem, then solve it.

a) I think of a number, multiply it by 3 and add 4. The answer is 19. What is the number?

b) A taxi charges \$4 plus \$3 for each kilometre. A trip costs \$25. How many kilometres long is the trip?

c) A rectangle is 6 cm wide and x cm long. Its perimeter is 38 cm. Find x.

d) Ana has 5 bags with the same number of marbles in each, and 12 loose marbles. She has 52 marbles altogether. How many marbles are in each bag?

★ Extension

7. Solve. There is an x on both sides.

a) $5x + 3 = 2x + 18$

b) $4x - 1 = x + 8$

c) $7x + 2 = 3x + 42$

d) $3x + 8 = 5x + 4$

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1-1 PRIVATE TUTORS

ALGEBRA

Solving Equations

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS equation *(whārite)* · solve *(whakaoti)* · solution *(otinga)* · inverse operation *(mahi kōaro)*

A One-step equations ●○○

Remember: Undo what has been done to x, and do the same to both sides. To undo $+7$, subtract 7. To undo $\times 4$, divide by 4.

1. Solve.

a) $x + 6 = 14 \quad x =$ _____ **b)** $x - 4 = 3 \quad x =$ _____ **c)** $3x = 27 \quad x =$ _____

d) $\frac{x}{4} = 5 \quad x =$ _____ **e)** $x + 8 = 5 \quad x =$ _____ **f)** $x - 3 = -9 \quad x =$ _____

g) $6x = -12 \quad x =$ _____ **h)** $\frac{x}{3} = -3 \quad x =$ _____ **i)** $2x = 7 \quad x =$ _____

j) $x + 2.5 = 6.5 \quad x =$ _____ **k)** $-4x = -12 \quad x =$ _____ **l)** $x - 1.5 = 1.5 \quad x =$ _____

B Two-step equations ●●○

Remember: Undo the adding or subtracting first, then undo the multiplying or dividing.

2. Solve. Show each step.

a) $2x + 5 = 11$ **b)** $3x - 4 = 11$ **c)** $4x + 1 = 25$ **d)** $5x + 8 = -2$

e) $6x - 5 = 19$ **f)** $2x + 7 = -1$ **g)** $4x + 9 = 9$ **h)** $\frac{x}{4} + 3 = 5$

i) $\frac{x}{2} - 3 = 4$ **j)** $17 - 3x = 5$ **k)** $10x - 9 = 21$ **l)** $7x + 4 = -3$

C Fraction and decimal answers ●●●

Watch out: The answer does not have to be a whole number. If $2x = 5$, then $x = \frac{5}{2}$, which is 2.5.

3. Solve. Give each answer as a fraction in its simplest form.

a) $3x + 2 = 7$ **b)** $2x + 7 = 12$ **c)** $5x - 1 = 2$ **d)** $4x + 1 = 8$

4. Solve. Give each answer as a decimal.

a) $2x + 3 = 10$ **b)** $4x + 1 = 11$ **c)** $5x - 1 = 6$ **d)** $10x + 7 = 16$

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D Checking a solution ●●○

Remember: Substitute the value into the equation. If both sides come to the same number, it is a solution.

5. Is the value of x a solution of the equation? Show your working.

a) $x = 4$ and $2x + 7 = 15$

b) $x = 6$ and $3x - 5 = 12$

c) $x = 3$ and $5x + 2 = 17$

d) $x = 12$ and $\frac{x}{3} + 2 = 6$

e) $x = 4$ and $5x - 6 = 3x + 2$

f) $x = -3$ and $12 - 2x = 6$

E Equations from words ●●●

6. Write an equation for each problem, then solve it.

a) I think of a number, multiply it by 4 and add 3. The answer is 27. What is the number?

b) A taxi charges \$5 plus \$2 for each kilometre. A trip costs \$23. How many kilometres long is the trip?

c) A rectangle is 5 cm wide and x cm long. Its perimeter is 38 cm. Find x.

d) Ana has 6 bags with the same number of marbles in each, and 15 loose marbles. She has 57 marbles altogether. How many marbles are in each bag?

★ Extension

7. Solve. There is an x on both sides.

a) $6x + 1 = 2x + 17$

b) $5x - 2 = 2x + 7$

c) $8x + 1 = 3x + 56$

d) $2x + 9 = 5x + 3$

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